

Exponentially Modified Gaussian Functions – a Good Model for Chromatographic Peaks in Isocratic HPLC?

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Key Words

Peak shapes in chromatography
Exponentially modified gaussian functions
Peak dispersion
Peak modelling

Summary

The Exponentially Modified Gaussian (EMG) function is fitted to a variety of typical peaks from isocratic HPLC, to assess its validity for peak characterization. This model has previously been applied, but with simplifying assumptions, allowing estimation of peak parameters from simple "front and rear" half-width measurements. The full EMG function is employed in this work. The results of the simplified approach have been compared to the EMG fitted results. Comparison is also made to fitted Gaussian functions. Results show that the EMG function is a good model for typical LC peaks, under a range of conditions, and that the simplified approaches give close correlation for a range of peak asymmetries.

Introduction

In normal chromatographic practice, an understanding of chromatographic peak shapes and the parameters characterizing them is useful for many reasons. These include the assessment of columns for comparison or monitoring purposes via the efficiency and asymmetry, or the design of HPLC systems and estimation of their compatibility with particular columns [1]. In more fundamental studies, aimed at deep understanding of the chromatographic processes, accurate characterization of peak shape is essential.

Frequently the model employed has been the Gaussian function. For some simple evaluations this is valid and is often selected for its simplicity and precision, but not its accuracy [2]. Dependent on the exact method of calculation, this assumption of Gaussian peak shapes can lead to errors in column efficiency, N , of over 100% [2], for skewed peaks.

An attractive method for peak characterization is statistical moments [2], as no peakshape is assumed. Statistical moments are summed across the peak between defined end-points, giving a total peak description. In theory, the method appears ideal, but in practice the positioning of these end-points particularly on the peak tail, causes large errors [3]. However it is used and recently manual measurements of second statistical moments gave some very elegant results for pure column dispersion measurements in OTLC [4].

A model that has some theoretical justification is the Exponentially Modified Gaussian (EMG) [5]. The gaussian contribution, characterised by σ , is due mainly to the column, although even theoretically the column contribution is not predicted to be perfectly symmetrical [6]. Dead volumes in the LC system produce an exponential modification to the peak [7], thus producing chromatographic peaks with variable degrees of tailing. Peaks of higher retention are generally more symmetrical than lower retention peaks as the external dispersion, mainly responsible for the tailing, is generally taken to be constant for all peaks in one run, but the column dispersion (symmetric component mainly), increases linearly with retention time [8]. In the course of this work it was found that τ , characterizing the exponential component was not independent of the retention.

Many studies have employed the EMG model [9, 10], obtaining useful results, without validation of the model for their sets of chromatographic peaks. The approximations introduced by Yau initially [11], later by Barber and Carr [12], and then Foley and Dorsey [13, 14] have made the EMG model accessible to chromatographers, allowing its simple application, without complex mathematical procedures.

The method of Yau was supported by good comparison of the calculated points of the EMG profile with the original profile [15]. These simplified approaches require measurement of the peak half-widths and full-widths at finite peak heights, which by assuming EMG functions, allow deconvolution of σ and τ , graphically [12] or by calculation [13, 14].

Although checking of the suitability of the EMG model for particular peaks is advocated [14], the method suggested is for consistency of results from measurements

at a range of percentage peak-heights. Another method is to check by using areas, and a recent study of 40 peaks [16] found that 90% fitted the EMG model within 5% by this criterion.

Although approximate EMG curves (employing up to 8 parameters [17]) have been fitted to LC peaks the proper EMG has only been used for deconvoluting unresolved GC peaks [18, 19], with no checking of the accuracy of the underlying model.

This paper describes our work curve-fitting EMG functions and Gaussian functions to a range of typical LC peaks, over 100 in total, via least squares fitting, and comparison of the resulting σ and τ values, with those calculated by the Barber and Carr method. The aims were to investigate the fitting of the EMG function directly, to determine if the EMG function is a good model and to compare the results from the simplified approaches. If this EMG model gives an accurate method for characterizing all chromatographic peaks, it could be applied universally, even as a facility within modern data-stations. This could provide accurate column efficiencies, accurate calculation of column and external dispersions, separation of the symmetrical and tailing components for fundamental studies, or used for constant monitoring of the condition of the column and system, aiding good chromatographic practice.

Method

Experimental

The chromatography was run on a quaternary liquid chromatograph, PU4100, (from Philips Scientific, Cambridge, UK), using manual injection. Data was collected on a DASH-16 12 bit A/D board (from METRABYTE, Mass. USA) fitted to an Olivetti M240 PC, running under ASYST, (from Keithley Instruments, Reading, UK), with a data collection interval of 60msec. The columns used were 10 and 22cm cartridges with 4.6mm i.d. packed with Spheri-5 RP-18, (from Brownless Labs., CA., USA).

The standard chromatographic conditions employed were 2ml/min, with 60:40 ACN/water at 40°C, with 5 μ l sample. Detection was at 243nm, with 60msec response time, at 0.2 AUFS. The sample contained 6 alkylphenones; acetophenone 7.584 μ g/ml, propiophenone 6.928 μ g/ml, butyrophenone 15.872 μ g/ml, valerophenone 19.808 μ g/ml, hexanophenone 12.290 μ g/ml, and octanophenone 18.400 μ g/ml, in acetonitrile:water 60:40. All variations in chromatographic conditions were relative to these standard conditions. Variations in temperature were from 40, to 50 and 60°C. Variations in eluents were 55:45 acetonitrile/water, 68:32 methanol/water and 42:58 tetrahydrofuran/water, designed to be approximately isoelutotropic. Variations in flow-rate were 0.5, 0.8, 1.0, 1.5, 2.0, 2.5ml/min. These variations were investigated individually, with all other parameters being kept constant at the standard conditions.

Mathematics

All data collection and analysis was carried out using the standard ASYST (ver 2.0) software package, which contains

a powerful set of fitting functions as well as comprehensive data collection facilities. The complete chromatogram was stored to disc and this was then split up into smaller files containing just one peak. These were then analysed, firstly by a computerised version of the Barber & Carr method [13] and then by the same method using recalculated data. The tables were recalculated because there were some typing errors in the values supplied by Barber & Carr and it was pointed out in a subsequent paper by Hanggi & Carr [20], that the EMG equation quoted in the original paper was incorrect. A comparison between the two sets of tables showed that the differences are small particularly compared with the typical errors found in the application of the method. The resulting values for tau, sigma and the peak time were fed into the curve fitting routines to provide good initial guesses, which were essential because of the iterative nature of the fitting processes. All the routines were used in the standard form as supplied with the ASYST package. The most useful iterative algorithm was a hybrid method that used the BFGS update as a Levenberg type modifier for the Gauss-Newton method (see the ASYST manual for further information). This algorithm, whilst it converges more slowly than the Newton method, is numerically more stable. The criterion for goodness of fit was least squares and the fitting was continued until the variation in all the fitted parameters was less than 0.0001. The number of data points used in the fitting varied between 200–1000 depending on the peak width. The fitting failed with peaks that were too nearly Gaussian or had too many data points (1000 was about the limit) due to the algorithms becoming numerically unstable. It was also possible to perform the fitting with a Gaussian function but this was only used on a few peaks because of the poor fits obtained.

The baseline was dealt with by being fitted (in a constant or linear form) with the EMG.

The normal form of the EMG function is

$$\text{EMG}(t) = \frac{1}{\tau \sigma (2\pi)^{1/2}} \int_0^t \exp\left(-\frac{(t-t_g-t')^2}{2\sigma^2}\right) \exp(-t'/\tau) dt' \quad (1)$$

Where t is the time, t_g is the centroid (maximum) of the Gaussian component, σ is the square root of the variance of the Gaussian component and τ is the time constant of the exponential function. This equation is not easily calculated but it can be transformed into a form which uses the erf function:

$$\text{EMG}(t) = \exp\left[\frac{1}{2}\left(\frac{\sigma}{\tau}\right)^2 - \frac{t-t_g}{\tau}\right] \left[\text{erf}\left(\frac{1}{2^{1/2}}\left[\frac{t_g}{\sigma} + \frac{\sigma}{\tau}\right]\right) + \text{erf}\left(\frac{1}{2^{1/2}}\left[\frac{t-t_g}{\sigma} - \frac{\sigma}{\tau}\right]\right) \right] \quad (2)$$

where

$$\text{erf}(z) = 2/(\pi^{1/2}) \int_0^z e^{-t^2} dt \quad (3)$$

The erf function is available in ASYST thus providing an easy way of calculating EMG functions.

Results and Discussion

A. Curve Fitting of EMG and Gaussian Functions

The least squares fitting of the EMG function is shown (Fig. 1), for peak 2 of chromatogram A, (Fig. 2A), run on 10cm cartridge column under standard conditions.

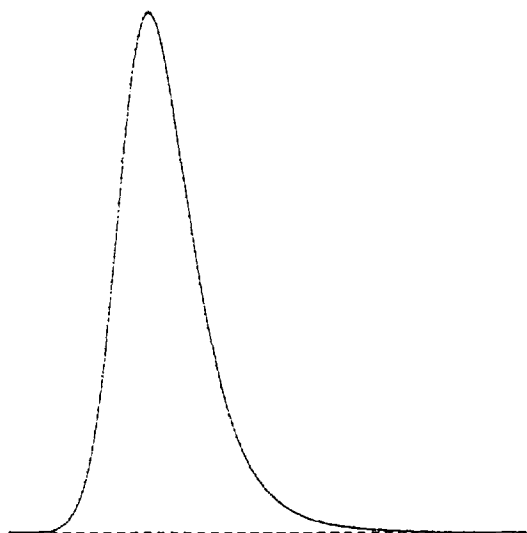


Fig. 1

EMG function fitted to the experimental data, for peak 2 of chromatogram A, (see Fig. 2) = fitted data, — = experimental data, using constant baseline.

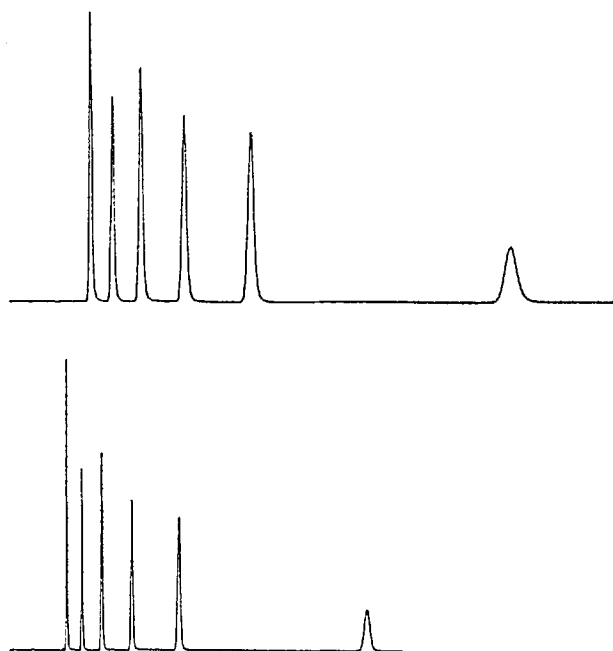


Fig. 2

Chromatograms under standard conditions on A. 10cm column, and B. 22cm column, with peaks number 1 to 6, eluting sequentially.

The close fitting right across the peak indicates the good fit of the EMG model to this peak. Results for peaks 1 to 6, of chromatogram A, are shown in Fig. 3, and the derived parameters listed in Table 1A. These peaks show a range of asymmetries (B/A) of 1.25 to 1.06, (measured at 10% peak height), or τ/σ 1.54 to 0.84. Similar results were obtained for peaks 1 to 6 of chromatogram B (Fig. 2B), on the 22cm cartridge column under standard conditions see Table 1B. Again the fit is good for these peaks, where the asymmetries range from 1.16 to 1.03, with τ/σ 1.206 to 0.346, (peaks 1 to 6 respectively).

Just for comparison, the fittings of Gaussian functions to peaks 1 and 6, of chromatogram A, are shown (Fig. 4). These peaks have high and low skew, ($B/A = 1.23$ and 1.06 respectively). Clearly large errors, particularly in the tail area, would be introduced if these typical chromatographic peaks were approximated to Gaussian functions, even for a peak of low skew, such as peak 6.

Thus for these peaks with a reasonably wide range of skewness, asymmetries 1.03 to 1.25 or τ/σ values 0.35 to 1.54, the EMG function appears to be a very good model. However it is not a perfect model, as shown by the small but significant mismatches. This occurs particularly at the top of early peaks, see peak 1, Fig. 3, and near the start and on part of the tail of some peaks. Difference plots, showing the difference between the experimental data and the fitted data, expressed as percentages of maximum peak-height, across these peaks reveal these points more clearly, Fig. 5. It is only by curve-fitting, overlaying and subtracting the curves that these small differences are revealed.

Clearly the derived τ values are not independent of retention, which according to our knowledge has not previously been observed. This suggests that τ contains a residual component of the column dispersion, consistent with theory predicting that column dispersion will have an asymmetrical component. However linear regression of the τ values to zero retention for the 10 and 22cm data does not give a consistent value.

Table 1. Listing the derived values from EMG fitted curves fitted to peaks from A. 10cm and B. 22cm columns.

Peak No.	t_g sec	σ sec	τ sec	B/A	τ/σ
A. 1	48.72	0.5102	0.7849	1.2301	1.5384
2	62.18	0.6275	0.8513	1.2524	1.3567
3	79.54	0.7882	0.9544	1.1897	1.2109
4	105.92	1.0313	1.1106	1.1512	1.0769
5	146.77	1.4095	1.3742	1.0963	0.9750
6	306.40	2.9649	2.4850	1.0613	0.8381
B. 1	98.00	0.6493	0.7830	1.1616	1.2059
2	124.53	0.8329	0.8351	1.1626	1.0026
3	158.88	1.1122	0.8774	1.0865	0.7889
4	211.31	1.4595	0.9901	1.1359	0.6784
5	293.81	2.0720	1.1455	1.0886	0.5528
6	622.77	4.5603	1.5796	1.0328	0.3464

B. Reproducibility of EMG Fitting

In Table II, are listed the results of deconvolutions on the first 5 peaks in 3 repeat runs, under the standard conditions on the 10cm column. The relative standard devia-

tions (RSD) are listed for t_g , τ , σ and B/A ratios. The RSD values of $<\sim 0.4\%$ for t_g and σ are typical of that obtained for retention data and thus this will be the cause of this variation. The variation in τ is greater, $<\sim 1.5\%$, due to injection technique and non-constant baselines.

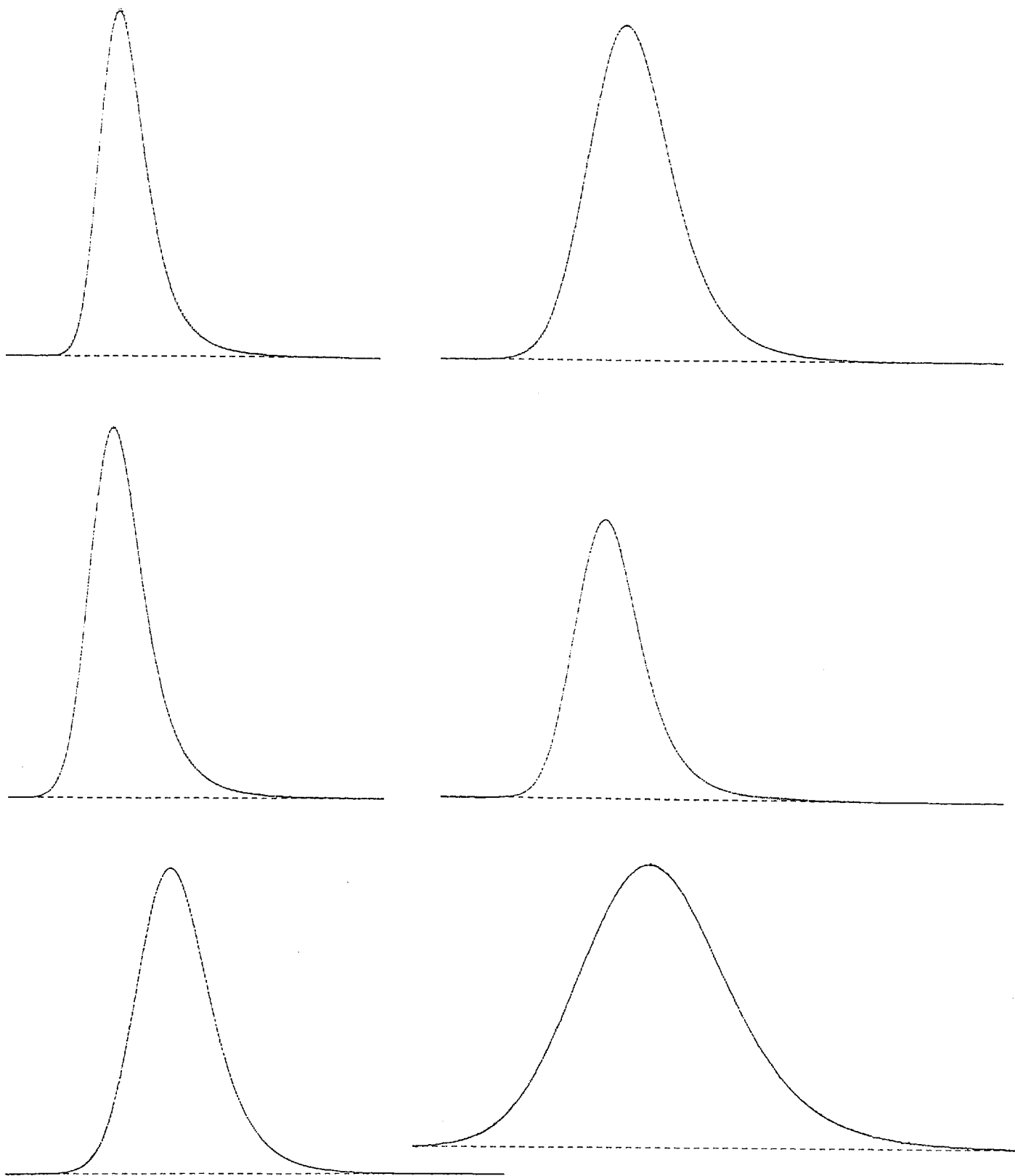


Fig. 3

EMG functions fitted to peaks 1 $\rightarrow$ 6, from 10cm column (see Fig. 2A), = fitted data, — = experimental data, all using constant baseline.

The larger variation in B/A ratios, RSD values upto 3.2%, will affect the precision of any calculations using these values. These errors are greater for the earlier peaks where the peaks are sharper. The graphical versions of the EMG calculation methods use these B/A values and thus will show a larger variation in their derived parameters (see later section).

C. Comparison of EMG Fitted Curves and Approximated Methods

As described earlier the Barber and Carr tables were recalculated using the corrected formulae. This allowed values

for t_g , σ and τ to be calculated from the A/B ratio at 10, 20, 30, 40 and 50% peak height and the retention time for each peak, thus giving 5 values for t_g , σ and τ for each peak. Initially we applied this method to a pure EMG function ($B/A = 1.3355$), to detect any variations in these derived values from an ideal EMG peak; Table III shows the results. The variations in the derived parameters are small, but the differences at different peak-heights point to the inherent inaccuracies of a graphical method. Clearly errors for chromatographic peaks cannot be less than this. As noted earlier the reproducibility measurements of B/A showed RSD values upto 3.2%, contributing to a lower precision on the derived values.

Table II. The reproducibility of EMG fitted curves to peaks from the 10cm column.

Peak No.	t_g sec	σ sec	τ sec	Fitted Area	B/A	Sq. Resid. [fitting]
1	48.7242	0.5102	0.7849	5037.6	1.2301	13385
1	48.5029	0.5130	0.8122	5065.5	1.3293	18997
1	48.4281	0.5122	0.8069	5365.8	1.2757	21461
RSD %	0.259	0.230	1.475		3.171	
2	62.1813	0.6275	0.8513	4185.9	1.2524	2434
2	61.8251	0.6286	0.8778	4210.5	1.3006	4095
2	61.7682	0.6273	0.8734	4447.4	1.3020	4148
RSD %	0.295	0.091	1.337		1.794	
3	79.5372	0.7882	0.9544	5707.0	1.1897	4624
3	78.9388	0.7844	0.9749	5721.8	1.2136	4240
3	78.9381	0.7846	0.9737	6045.0	1.2096	4434
RSD %	0.357	0.222	0.971		0.868	
4	105.9184	1.0313	1.1106	5714.7	1.1512	1113
4	104.9321	1.0272	1.1315	5728.4	1.1310	280
4	105.0208	1.0267	1.1372	6052.0	1.1800	535
RSD %	0.423	0.200	1.015		1.742	
5	146.7715	1.4095	1.3742	6884.9	1.0963	2699
5	145.3754	1.4070	1.4040	6902.5	1.0991	1572
5	145.6358	1.4079	1.4161	7286.5	1.1265	1549
RSD %	0.415	0.073	1.259		1.230	
* 6	306.3997	2.9649	2.4850	4499.3	1.0613	823
* 6	305.1588	2.954	2.5656	4759.6	1.1631	782

* only 2 runs available

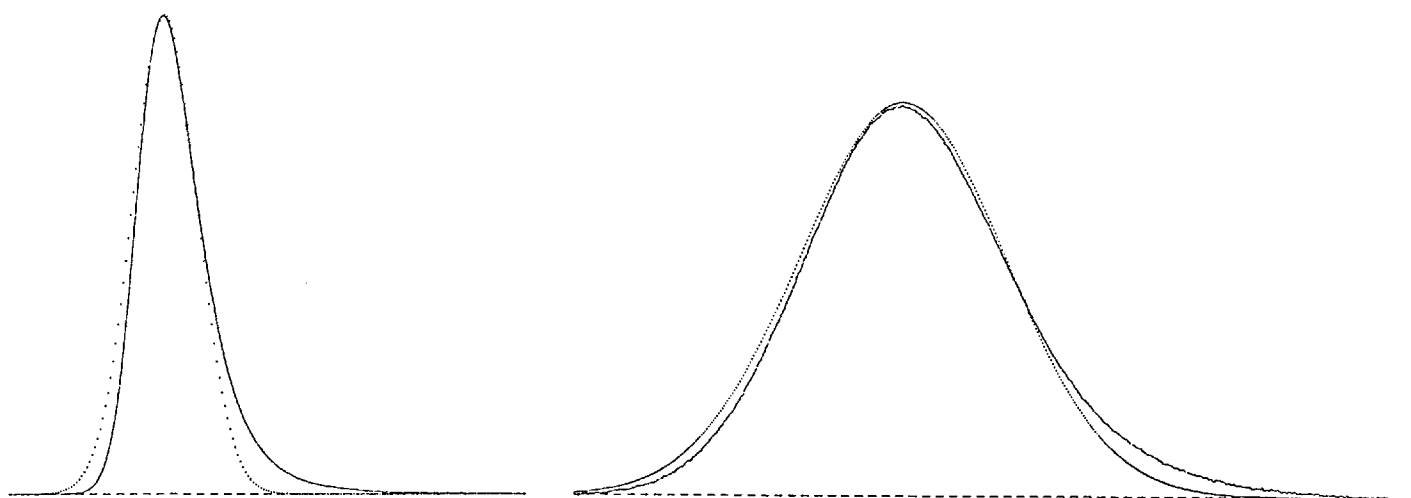


Fig. 4

Gaussian functions fitted to peaks 1 and 6 of chromatogram A, of Fig. 2A.

Table III. Derived values from B/A ratios at various peak heights, for pure EMG function with $t_g = 300$ sec, $\sigma = 2.6$ sec and $\tau = 3.5$ sec.

Peak Height Ratio	σ sec	τ sec	t_g sec
0.1	2.5958	3.5066	299.9895
0.2	2.5939	3.5116	299.9886
0.3	2.5926	3.5167	299.9874
0.4	2.5904	3.5259	299.9852
0.5	2.5875	3.5394	299.9818

Table IV. Comparison of derived values from A. approximated method and B. EMG fittings.

Peak No.	Pk. Height	t_g sec	σ sec	τ sec	Comments
A. 6	0.1	306.5199	3.0361	2.4065	B/A
6	0.2	306.5860	3.0509	2.2747	= 1.0613
6	0.3	306.6411	3.0657	2.1693	*
6	0.4	306.7331	3.1074	1.9945	
6	0.5	306.8074	3.1311	1.8660	
5	0.1	146.8222	1.4439	1.3346	B/A
5	0.2	146.8435	1.4485	1.2867	= 1.0962
5	0.3	146.8646	1.4542	1.2386	*
5	0.4	146.8843	1.4620	1.1964	
5	0.5	146.9163	1.4756	1.1294	
4	0.1	105.9293	1.0420	1.0986	B/A
4	0.2	105.9315	1.0372	1.0967	= 1.1525
4	0.3	105.9349	1.0393	1.0870	*
4	0.4	105.9402	1.0409	1.0737	
4	0.5	105.9448	1.0431	1.0615	
3	0.1	79.5469	0.7940	0.9403	B/A
3	0.2	79.5477	0.7931	0.9390	= 1.1891
3	0.3	79.5467	0.7924	0.9422	*
3	0.4	79.5475	0.7940	0.9388	
3	0.5	79.5485	0.7972	0.9333	
2	0.1	62.1796	0.6216	0.8488	B/A
2	0.2	62.1766	0.6218	0.8577	= 1.2522
2	0.3	62.1734	0.6219	0.8672	*
2	0.4	62.1691	0.6217	0.8801	
2	0.5	62.1670	0.6229	0.8852	
1	0.1	48.7513	0.5152	0.7566	B/A
1	0.2	48.7492	0.5204	0.7562	= 1.2300
1	0.3	48.7464	0.5252	0.7590	*
1	0.4	48.7475	0.5312	0.7487	
1	0.5	48.7525	0.5412	0.7225	
EMG Fitted Values		t_g	σ	τ	
B. 6		306.40	2.9649	2.4850	
5		146.77	1.4095	1.3742	
4		105.92	1.0313	1.1106	
3		79.54	0.7882	0.9544	
2		62.18	0.6275	0.8513	
1		48.72	0.5102	0.7849	

* from 0.1 peak height measurements.

The parameters derived from our recalculated Barber and Carr approach, at various peak heights for the 10cm column data (see Table IV), are compared with the same parameters derived from the EMG fitting. The percent differences of these σ and τ values from those derived from the EMG fittings are plotted against t_g , see Fig. 6 and 7. Several points are evident. The σ and τ values derived from the 10% peak height measurements show the lowest

Table V. EMG Derived values for various temperatures.

Temp °C	Peak No.	t_g sec	σ sec	τ sec	τ/σ
40	1	48.4281	0.5123	0.8067	1.5747
40	2	61.7682	0.6273	0.8734	1.3923
40	3	78.9381	0.7846	0.9737	1.2410
40	4	105.0208	1.0267	1.1372	1.1076
40	5	145.6358	1.4079	1.4161	1.0058
50	1	47.4522	0.4846	0.8022	1.6554
50	2	59.9946	0.5901	0.8667	1.4687
50	3	76.0361	0.7345	0.9608	1.3081
50	4	100.1148	0.9519	1.0976	1.1531
50	5	137.0382	1.2967	1.3313	1.0267
60	1	45.8212	0.4629	0.7521	1.6248
60	2	57.1519	0.5619	0.8054	1.4334
60	3	71.5132	0.6953	0.8818	1.2682
60	4	92.829	0.8987	0.9819	1.0926
60	5	125.1787	1.2171	1.1402	0.9368

Table VI. EMG fitted values for approximately isoelutotropic solvents. [The column pressure, P , is indicative of the solvent viscosity.]

Eluent	t_g sec	σ sec	τ sec	Sq. Resid.	P (bar)
ACN/H ₂ O AT 55/45	53.3793	0.5714	0.8409	22972	72
	71.2800	0.7305	0.9763	5494	
	95.2369	0.9541	1.1551	3258	
	132.9101	1.3096	1.4464	947	
MeOH/H ₂ O AT 68/32	193.2973	1.8973	1.9335	1048	122
	51.4481	0.6503	1.0627	25671	
	67.3110	0.8182	1.1216	6201	
	90.2047	1.0717	1.2419	5227	
	130.1149	1.4978	1.4992	3296	
THF/H ₂ O AT 42/58	197.9665	2.2189	1.9745	670	154
	61.7380	0.7245	1.0516	64409	
	88.6256	1.0105	1.2132	1227	
	120.6969	1.3713	1.4315	616	
	166.4267	1.886	1.7387	648	
	229.7328	2.6337	2.1941	749	

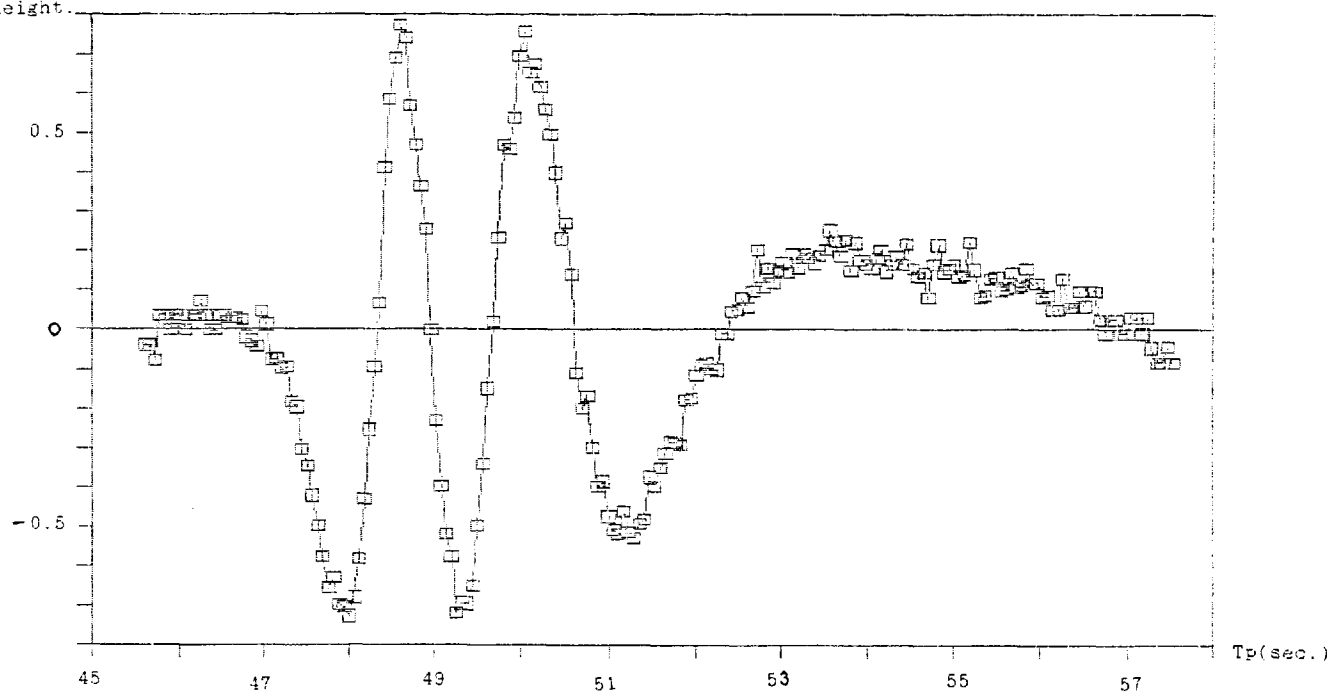
difference (except for σ value for peak 4). At higher t_g values (peaks 5 and 6, where B/A is low, 1.096 and 1.061 respectively), the derived σ and τ values diverge greatly. According to Barber and Carr, when $B/A > 1.06$, then σ and τ can be found with reasonable accuracy and precision, typically $\sigma \pm 3\%$, $\tau \pm 5\%$. However our results for peak 5 show a large spread, although the peak appears to fit an EMG function well. This may be due to a t_g effect.

For peaks 2, 3 and 4, (B/A 1.252 to 1.153), the percent differences of all derived values from the EMG fitted values are within 1.2% for σ , and 4.2% for τ , and specifically for the 10% values σ and τ are within +1% and -2%. For peak 1, the fit to an EMG function is less good, and this is reflected in the derived values varying greatly, σ 1 to 6%, and τ 3.5 to 8%.

D. Observation of Significant Effects on Peak Shape from Normal Chromatographic Conditions

Temperature Variations: The variations of the deconvoluted parameters with temperature for the first 5 peaks, run under standard conditions on the 10cm column, but

Percent
difference
between
experimental
points &
fitted EMG,
relative to
maximum peak
height.



Percent
difference
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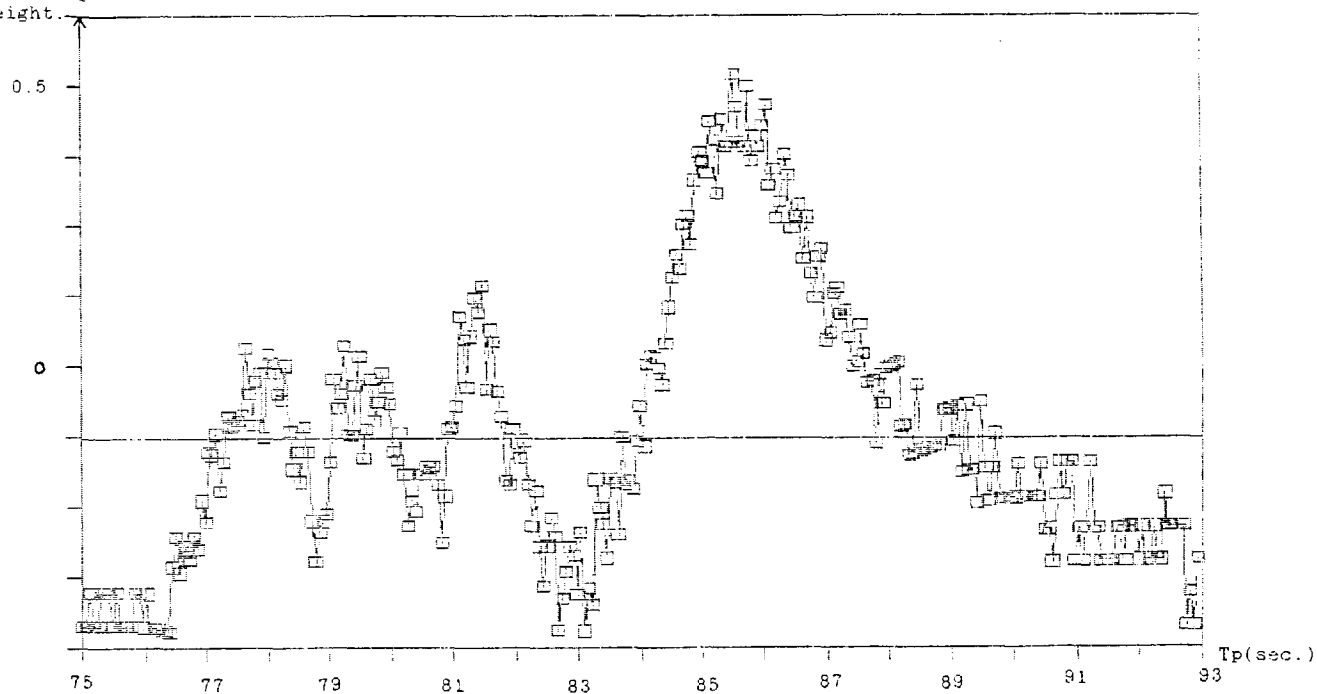


Fig. 5

Plot of difference between experimental and fitted data across peaks 1 and 3, of Fig. 2A, expressed as a percentage of maximum peak height.

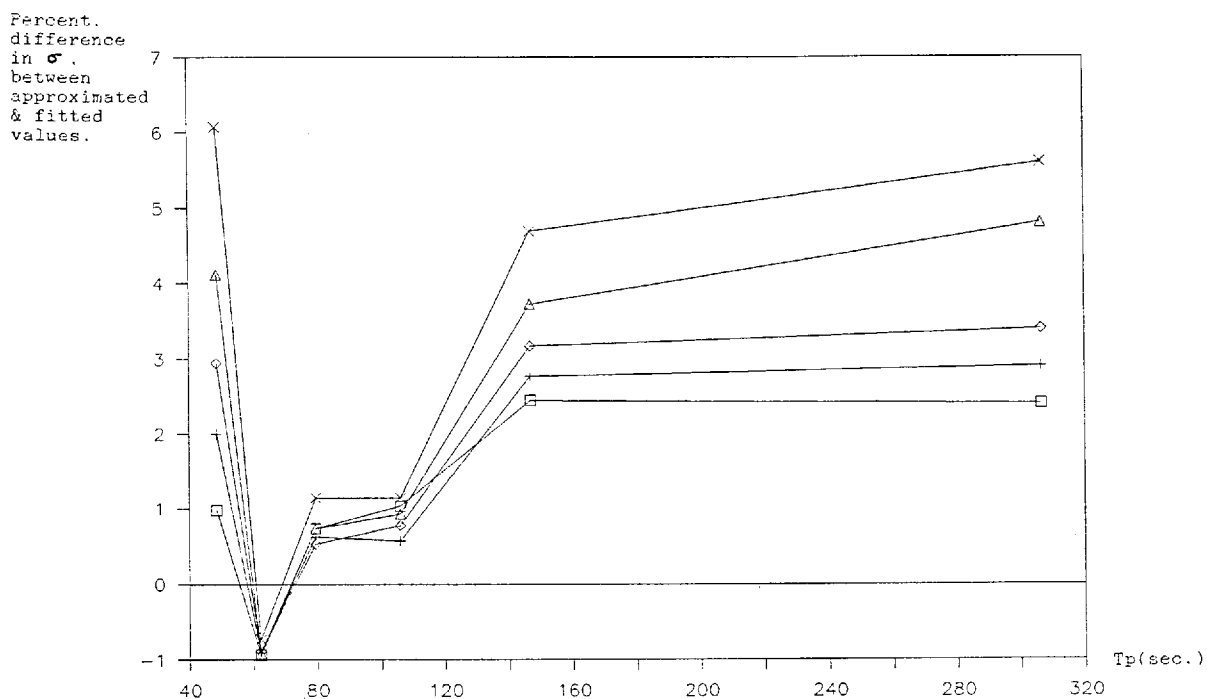


Fig. 6

Percent difference in σ , for values derived from the 5 peak ratios compared to the curve fitted value for each of 6 peaks, plotted against t_g , for 10cm column under standard conditions, see Table IV. $\square = 0.1$, $+$ = 0.2, $\diamond = 0.3$, $\Delta = 0.4$, $\times = 0.5$ peak height ratios.

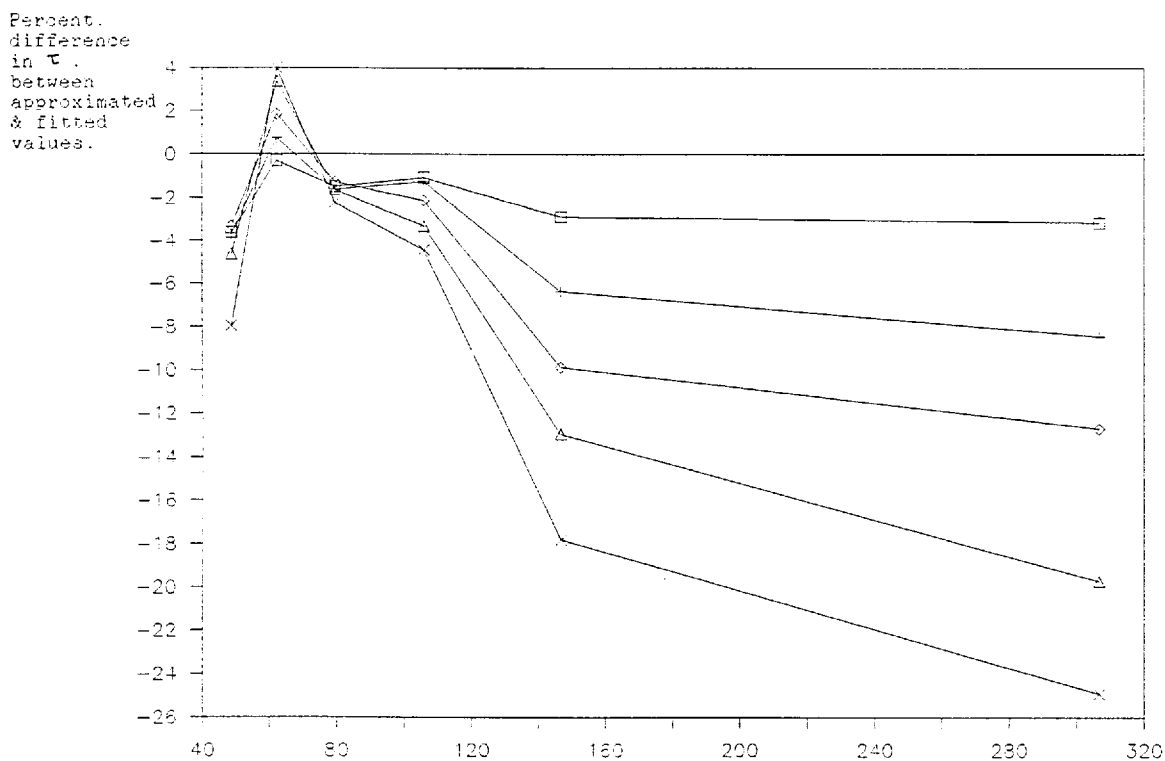


Fig. 7

Percent difference in τ , for values derived from the 5 peak height ratios compared to the curve fitted value for each of 6 peaks, plotted against t_g , for 10cm column under standard conditions, see Table IV. $\square = 0.1$, $+$ = 0.2, $\diamond = 0.3$, $\Delta = 0.4$, $\times = 0.5$ peak height ratios.

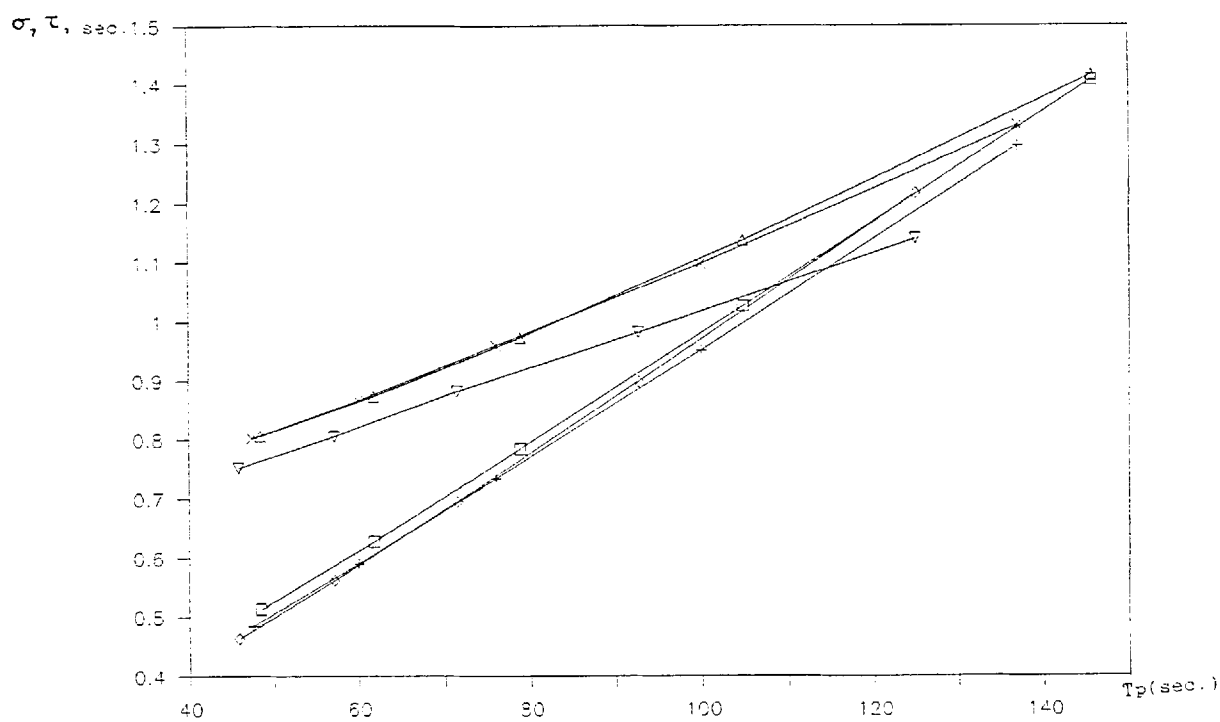


Fig. 8

Plot of σ and τ values, derived from EMG fitting of the six peaks run on 10cm column under standard conditions, but at 40°C $\square, \Delta$ (σ and τ respectively), 50°C $+, \times$, and 60°C $\diamond, \nabla$, against T_p , see Table V.

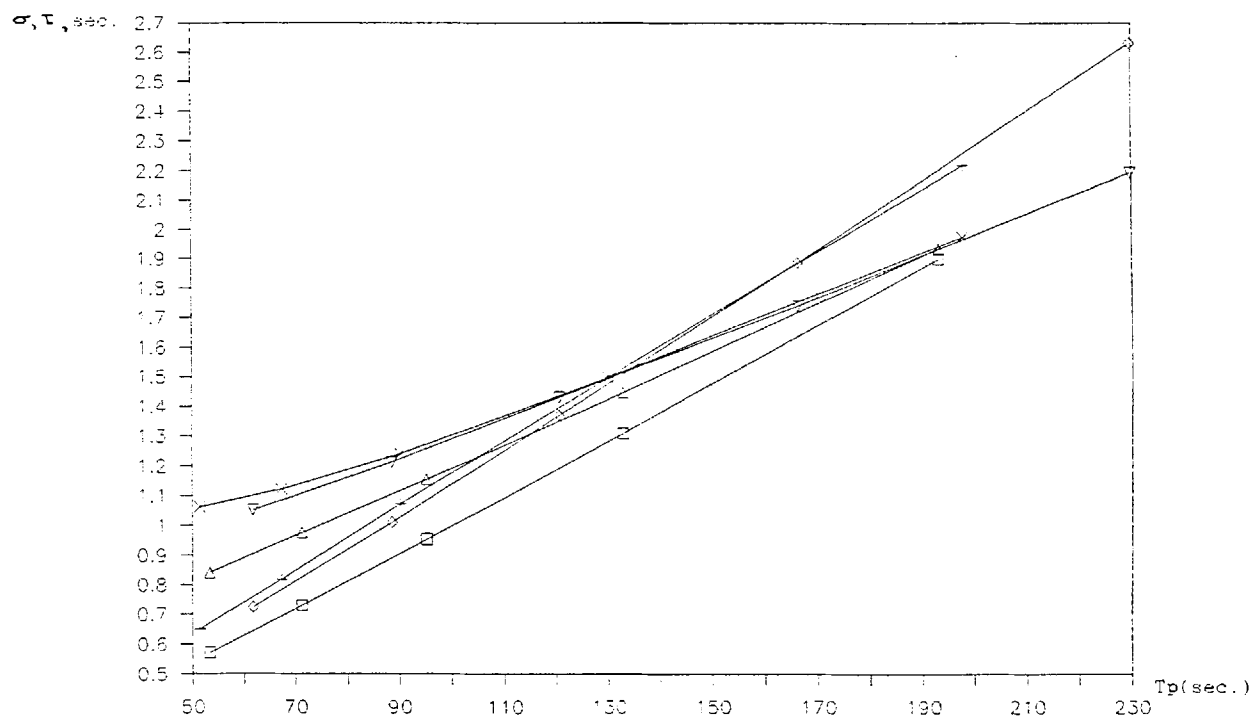


Fig. 9

Plot of σ and τ values, derived from EMG fitting of 6 peaks run on 10cm column under standard conditions, but at 55:45 acetonitrile/water $\square, \Delta$, (σ and τ respectively), 68:32 methanol/water $+, \times$, 42:58 THF/water $\diamond, \nabla$, see Table VI. [Column back pressures can be used as an indicator of the relative viscosities of the eluents.]

at 40, 50 and 60°C, are listed in Table VI, and shown in Fig. 9. Clearly there are different effects on σ and τ due to temperature changes, pointing to the different mechanisms controlling σ and τ . The drop in σ with increasing temperature is in parallel with the drop in retention times, so that the plot of σ against t_g for each temperature are almost coincident. However the drop in τ with increasing temperature is very different, with larger changes occurring between 50 and 60°C. Fundamental investigations are needed to understand these variations.

Eluents: The effects on σ and τ of common eluent mixtures, that are approximately isoeluotropic, were investigated for the first 5 peaks, run under standard conditions, except that solvent compositions used were 55:45 acetonitrile/water, 68:32 methanol/water, and 42:58 tetrahydrofuran/water, (see Table VII, Fig. 9). Clearly behaviour of σ and τ under these changes of eluent are very different, with the lower viscosity solvents acetonitrile/water behaving very differently to the higher viscosity solvents methanol/water and tetrahydrofuran/water. Acetonitrile/water thus has beneficial effects on σ and τ , and will give higher efficiency separations.

Flow-Rates: The effects on σ and τ of variations in flow-rates were studied, across the flow-rate range of 0.5 to 2.5 ml/min. The values derived from the fitted EMG functions, for the 6 peaks at each flow-rate, are shown (see Table VII, and Fig. 10, 11). The values are listed in volume units, thus allowing for the variations in flow-rate. Clearly σ has a minimum value for each peak around 25 μ l/sec (1.5 ml/min), indicative of the minimum in the H versus flow plots. The dependence of τ on flow has very different behaviour, decreasing with increasing flow-rate.

Conclusions

The EMG model is a good model for isocratic LC peaks, but it is not perfect; — the greatest deviations being found for early peaks with high asymmetries. The fitting procedure described here worked well, allowing good peak shape description. This allows derivation of peak parameters at lower asymmetries than the Barber and Carr type approach. The reproducibility on the 5 test peaks was acceptable, (RSD for $\sigma < 0.42\%$, and $\tau < 1.48\%$).

Table VII. EMG fitted values from various flow-rates.

Peak No.	t_g sec	sq. res.	Flow μ l/s	σ μ l	τ μ l	V_g μ l
1	189.77	74581	8.3333	17.2833	33.6767	1581.42
2	240.07	32012	8.3333	21.4342	37.7925	2000.58
3	304.87	60438	8.3333	27.0150	43.2992	2540.58
4	405.22	41204	8.3333	36.8158	52.8883	3376.83
5	567.38	34584	8.3333	52.9767	67.2308	4728.17
6	1219.48	4585	8.3333	121.8500	125.6775	10162.33
1	122.05	104882	13.3333	17.0040	35.1907	1627.33
2	156.36	42529	13.3333	21.0800	39.1373	2084.80
3	201.14	58678	13.3333	26.2160	44.0253	2681.87
4	268.64	26895	13.3333	34.0627	51.6013	3581.87
5	372.89	19459	13.3333	46.9987	63.8560	4971.87
6	775.12	4893	13.3333	102.4480	115.0053	10334.93
1	97.30	87048	16.6667	16.8517	30.9317	1621.67
2	124.07	31103	16.6667	20.5067	33.9033	2067.83
3	158.60	34253	16.6667	25.3100	37.9850	2643.33
4	211.03	11459	16.6667	32.8900	45.1817	3517.17
5	292.45	5216	16.6667	45.1417	56.9050	4874.17
6	610.55	1073	16.6667	96.6483	107.9067	10175.83
1	64.34	54247	25.0000	16.7625	26.6000	1608.50
2	81.88	13973	25.0000	20.2100	29.5550	2047.00
3	104.38	18263	25.0000	24.8325	34.0600	2609.50
4	138.58	2881	25.0000	32.2700	40.9700	3464.50
5	191.86	2804	25.0000	44.0750	52.2425	4796.50
6	400.17	812	25.0000	93.7150	98.0050	10004.25
1	48.42	21461	33.3333	17.0733	26.8967	1614.00
2	61.77	4148	33.3333	20.9100	29.1133	2059.00
3	78.94	4434	33.3333	26.1533	32.4567	2631.33
4	105.02	534	33.3333	34.2233	37.9067	3500.67
5	145.64	1549	33.3333	46.9300	47.2033	4854.67
6	305.16	782	33.3333	98.4567	85.5200	10172.00
1	39.29	7867	41.6667	17.5667	24.8583	1637.21
2	50.32	1829	41.6667	21.8292	27.2167	2096.67
3	64.55	5898	41.6667	27.6958	30.6958	2689.58
4	86.31	1726	41.6667	36.7833	35.8125	3596.38
5	120.39	3440	41.6667	50.9958	44.5250	5016.13
6	254.50	1245	41.6667	107.2917	81.5542	10604.04

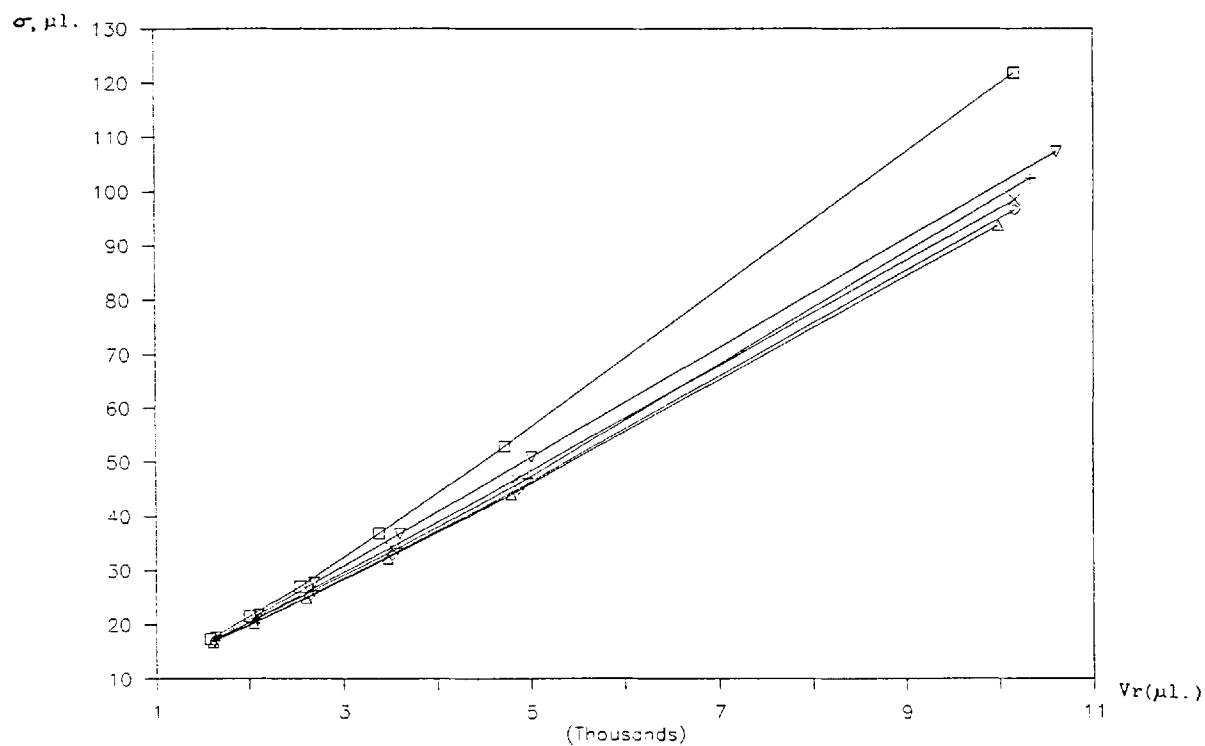


Fig. 10

Plot of σ values, derived from EMG fitting of 6 peaks run on 10cm column under standard conditions, but at flow-rates of $\square = 8.33$, $+$ = 13.33, $\diamond = 16.67$, $\triangle = 25.00$, $\times = 33.33$, $\nabla = 41.67 \mu\text{l}/\text{min}$ (see Table VII).

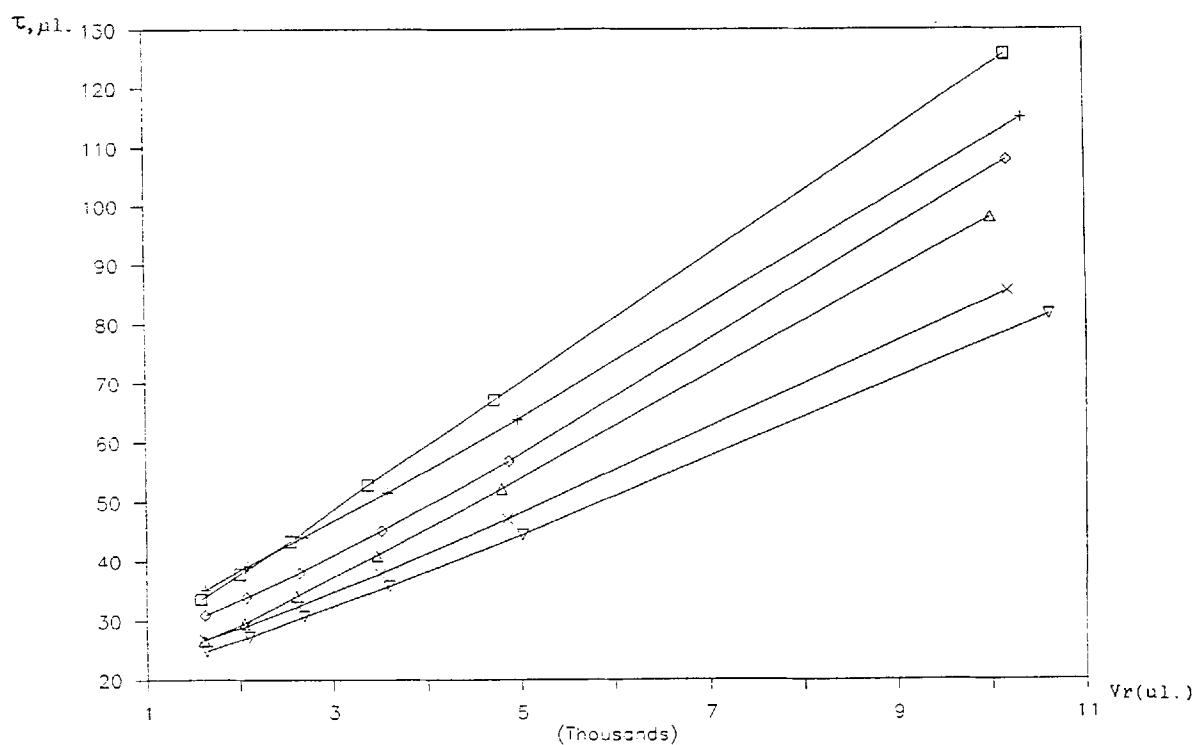


Fig. 11

Plot of τ values, derived from EMG fitting of 6 peaks run on 10cm column under standard conditions, but at flow-rates of $\square = 8.33$, $+$ = 13.33, $\diamond = 16.67$, $\triangle = 25.00$, $\times = 33.33$, $\nabla = 41.67 \mu\text{l}/\text{min}$ (see Table VII).

With the approximated method of Barber and Carr the 10% peak height values gave the closest agreement to the EMG fitted values, for all peaks. For different peak height ratios, the derived values agreed reasonably ($\sigma \sim 1\%$, $\tau \sim 4\%$), except at low asymmetries ($B/A = 1.06$ and 1.10), and where the EMG fit is not good. The Barber and Carr approaches can be useful for many peaks, with a spread of results from different peak height ratios indicative of a problem. However it is limited to peaks of significant asymmetry ($B/A > 1.1$ from these results).

The variations in peak shape observed for normal chromatographic conditions, show that the deconvoluted σ and τ values provide a handle for studying these effects. Clearly assigning σ to column processes and τ to external processes exclusively is an oversimplification, as evidenced here by the dependence of τ on retention. However the differing behaviour of σ and τ with eluents, temperature and flow-rates show there is a definite distinction in the origin of these components. An understanding of these deconvoluted parameters is needed for interpretation of results.

The EMG function is a good model for chromatographic peaks and fitting appears to provide an accurate method for obtaining peak shape parameters for whatever purpose: — fundamental studies, column and system studies or routine column efficiency monitoring.

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